Three log companions + Chebyshev = a 2500-year problem closed.

13 constants	14 constants	vs. AAA	vs. prior author record
3.3 ppt	$\mathbf{2.1 \times 10^{-13}}$	**147,000**×	**2**× on half the params

The Teardown: 2500 years of ellipse-perimeter approximation

Method	Params	Peak error	At $b \to 0$
Ramanujan R1 / R2 (1914)	0	$\sim$ 0.1–0.3 ppm	plateaus
Cantrell (2001–04)	few	$\sim$ 1 ppm	slow decay
Villarino (2005)	0	$\delta_5 = 3/2^{17}$	analytic
Koshy / Moscato (2024)	5–8	$<$ 3 ppm	interpolatory
Ayala-Raggi F_2–F_5 (2025–26)	$\sim$15	0.0216 ppm	series-truncated
AAA (Trefethen et al., 2018)	30	$\sim 10^{-13}$	*cannot represent log*
SRIRACHA (Mamoun, prior record)	$\sim$25	7 ppt	log-branch-aware (gate)
UNDERWOOD F3 (this work)	**13**	**3.3 ppt**	*flat to floor*
UNDERWOOD F4 + Remez	**14**	$\mathbf{2.1 \times 10^{-13}}$	*machine* ε

All rows verified at 50-digit `mpmath` precision. The green rows are the only ones whose error is *flat* (not plateauing) as $b \to 0$. Separation theorem in paper §8: gap vs. any rational method grows without bound.

Paper + code + Lean

`github.com/monir/math`

Reproduce in 5 seconds:

```
python verify_all_claims_v2.py
lake build Logatoms
```

77-page paper

`paper_mamoun_ellipse_underwood_two_foci_one_cup_v2_19-April-2026.pdf`

The one-line idea. The ellipse perimeter has a logarithmic branch point at the degenerate limit $b \to 0$. Every prior method fit *smooth* functions to a *non-smooth* target. UNDERWOOD writes the log modes explicitly: three named log companions $\xi^{2+j}\log(1/\xi)$ added to a Chebyshev basis, QR solve, done. Residual is analytic; Chebyshev owns it at geometric rate; the gap to any rational method grows without bound.

Verification seal. ✓ Every numerical claim reproducible at 50 digits. ✓ Separation theorem proved (super-geometric envelope vs. rational impossibility). ✓ Mathematical substrate (the λ-ladder of logatom primitives) formalised in Lean 4 + Mathlib v4.28.0: **4 modules, 9 theorems, fully sorry-free**. ✓ 77-page paper with 20 open-question "landings" table engaging adjacent literature (AAA, Gopal-Trefethen, XFEM crack-tip enrichment, Stahl 1997, Gonchar-Rakhmanov 1989, Rokhlin quadrature, and more).